LESSON 7.4

INTERPRETING CIRCLE GRAPHS

LESSON INTRODUCTION

MA.7.DP.1.4 Use proportional reasoning to construct, display, and interpret data in circle graphs.

LEARNING TARGETS

- I can interpret the data in a circle graph.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.9.DP.1.1 Given a set of data, select an appropriate method to represent the data depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

ESSENTIAL QUESTIONS

- How can you use percentages given in a circle graph to interpret the data represented in the graph?
- How can you use given data to determine the angles and percentages the data represents in a circle graph?
- How do you construct a circle graph?

LESSON NARRATIVE

This lesson is a culmination of the previous three lessons. Students calculate percentages and angle measures given raw data and use them to construct circle graphs that represent the data. Students interpret the data in circle graphs.

COMPONENT SUMMARY

7.4.1 WARM-UP *(~5 MIN.)*

Would you rather? using pizza slice sizes

7.4.2 GUIDED INSTRUCTION *(10 MIN.)*

Interpret data in circle graph

7.4.3 COLLABORATION *(20 MIN.)*

Collaboratively construct and interpret circle graphs

7.4.4 YOUR TURN! *(5 MIN.)*

Interpret data in circle graph

7.4.5 WRAP-UP *(~5 MIN.)*

Reflect on mastery of lesson learning target

RESOURCES

GLOSSARY

Key Terms:

- N/A

Additional Terms:

- N/A

MATERIALS

- Protractor
- Mini whiteboards
- Dry-erase markers
- (optional) Chart paper & markers

ADDITIONAL PREPARATION

- Copies of recipes for partners. Partners should not share the recipe lists with each other.

Partner A: Sweet & Salty	**Partner B: Party Ranch**
6.5 cups of popcorn	4 cups square wheat cereal
2 cups of frosted oat and marshmallow cereal	4 cups square rice cereal
1.5 cups of mini pretzels	2 cups mini pretzels
1 cup of chocolate chips	1.5 cups corn chips
1 cup butterscotch chips	1.5 cups cheddar crackers

- Copies of a blank circle with the center point given.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students might not measure from the last ray of the previous angle to measure the next consecutive sector when constructing circle graphs.

THINGS TO CONSIDER

- Consider creating and displaying an anchor chart with the steps needed to construct a circle graph. Include an example and visuals.
- Consider creating larger circle templates.
- Allowing students to use calculators during the lesson will help students focus on discovery and process of finding central angles and percentages when constructing circle graphs.

LITERACY AND LANGUAGE ROUTINES

Poll the Class

In 7.4.2 Guided Instruction, use this routine to get a snapshot of what students already know about interpreting circle graphs. This routine offers a low-risk approach to engage students in sharing their initial interpretations of the data.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.6.1 Reasonableness

Students have opportunities to construct and check the accuracy of circle graphs that represent real-world data throughout the lesson. Remind students to informally check their answers by asking themselves if they make sense and formally check their answers by determining if the sum of percentages equal 100% and sum of angle measures equal 360°.

DIFFERENTIATE INSTRUCTION

7.4.2 Guided Instruction offers a scaffolded algorithm for determining the percentages of the data. If needed, students can use this scaffolded algorithm throughout the lesson. In 7.4.3 Collaboration, consider strategically assigning the recipes. Recipe A has fewer calculations, as two of the ingredients have the same measure in the recipe. Also, consider providing students with tables with some of the ingredients and calculations entered as an additional scaffold as needed.

7.4.1 WARM-UP: WOULD YOU RATHER?

TEACHER MOVES

Some students may reply that they would choose one pizza over the other based on the images (i.e. the toppings in the image or which pizza appears larger). To engage students in mathematical thinking, consider asking questions, such as:

- If you could have any toppings on either pizza, would that change your decision? (In response to students who chose based on toppings) Why?
- How do you know that your slice from the _____ pizza will be larger/smaller?
- How could you determine the area of each pizza? How could you use this to determine the area of each slice? • What vocabulary word can be used to refer to a slice of pizza?

TEACHER MOVES

Consider that some students may not like pizza, so they might choose the smaller slice and justify that mathematically. There also may be religious reasons why a student might select one over the other based on the toppings. Asking the first question listed above could address this potential constraint for students.

If a student doesn't like pizza, consider changing the context to pies or circular cookie cakes.

7.4.1 Warm-Up: Would You Rather?

A 10-inch pizza with 8 slices and a 14-inch pizza with 10 slices were ordered for a party.

$10''$ **Pizza**	$14''$ **Pizza**
	

1. Would you rather have a slice from the $10''$ pizza, or a slice from the $14''$ pizza?
I would rather have 1 of 10 slices from the $14''$ pizza because the pizza is bigger.

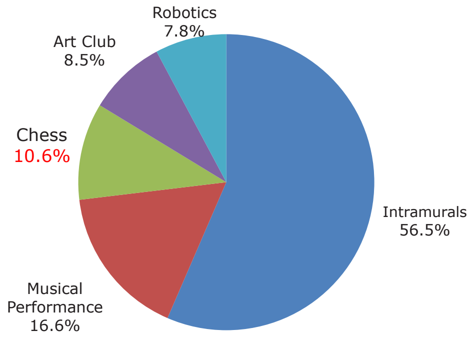
2. Explain your choice or justify it by showing your work.
I chose the second pizza because the area per slice of the first pizza is $9.8\ in^2$ while the area per slice of the second pizza is $15.386\ in^2$.

7.4.2 Guided Instruction: Interpreting a Circle Graph

1. The graph reflects the top five after-school activities at the Capital City Recreation Center based on the number of participants for each activity.

   a. Fill in the missing percent for Chess Club.

Top 5 After-School Activities



   b. Which activity has the most participants?
   Intramurals

   c. Which activity has the least participants?
   Robotics

   d. In which activity do about ½ of the kids participate?
   Intramurals

   e. In which two activities, when combined, do about $\frac{1}{4}$ of the kids participate?
   Art Club and Music Performance OR Chess and Music Performance

   f. Discuss with your partner how the circle graph helped you to answer the questions.
   Answers will vary. I was able to use the circle graph to compare the sectors to each other. I was able to visualize which percentages were closest to the fraction amounts.

2. The Budget Committee members at the Capital City Recreation Center want to look at ways to better support their after-school programs. They decided to purchase one sketch book for every student in the art club.

   a. Determine how many of the 283 total students that participate in extracurricular activities are in the art club by filling in the blanks.

Find 8.5% of 283.	8.5% of 283 ⟶ 8.5% $\boxed{\cdot}$ 283 8.5% $= \frac{8.5}{100} =$ $\boxed{.085}$ $\boxed{.085} \cdot 283 =$ $\boxed{24}$ students in the art club

   b. Determine how many of the 283 total students that participate in extracurricular activities are in the robotics club.

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7.4.2 GUIDED INSTRUCTION: INTERPRETING A CIRCLE GRAPH

TEACHER MOVES

Poll the Class

Consider giving students mini whiteboards to use when responding to questions 1a-1e. Use this routine to get a snapshot of what students already know about interpreting circle graphs. This routine offers a low-risk approach to engage students in sharing their initial interpretations of the data.

TEACHER MOVES

Incorporating the Desmos scientific calculator can add a powerful tool for students who struggle with completing steps on a handheld calculator. The Desmos calculator allows students to input complicated arithmetic that would require several steps on a handheld calculator. This eliminates common careless errors that many students make.

For example, students could simply enter $\frac{8.5}{100} \times 283$ in their calculator to determine the number of students in art club.

7.4.3 COLLABORATION: WHAT DOES THE GRAPH SAY?

DIFFERENTIATE INSTRUCTION

Consider strategically assigning the recipes. Recipe A has fewer calculations, as two of the ingredients have the same measure in the recipe. Also, consider providing students with tables with some of the ingredients and calculations entered to reduce the number of calculations needed and provide a model for students to reference.

TEACHER MOVES

For question 2, when a group is finished, consider allowing them compare to their results with another group which was assigned the same recipe, and task the students with coming up with a consensus for all of the answers.

ENRICHMENT

Consider fostering a discussion about converting fractions to decimals, rounding, and accuracy.

- When is a decimal an exact representation of its fractional equivalent?
- What place value would it make sense to round to in this component?
- Can you think of a situation where it would make sense to round to a different place value?

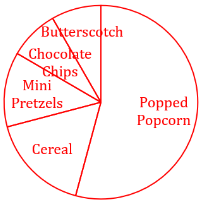
7.4.3 Collaboration: What Does the Graph Say?

Determine with your partner who will be partner A and who will be partner B. Partners A and B will be given two different ingredient lists for two popular holiday snack mixes and a blank circle. Do not let your partner see your list.

1. Use your recipe to fill in the chart. Then, construct a circle graph for your recipe using the circle provided. Round each percent to the nearest tenth and angle measure to the nearest whole number.
   Answers for both partners are given.

Your Recipe Salty & Sweet Mix
Total Number of Cups 12

Ingredient	ingredient (cups) / total cups	Percentage	Angle Measure (°)
Popped Popcorn	$\frac{6.5}{12}$	54.2%	195°
Frosted Oat & Marshmallow Cereal	$\frac{2}{12}$	16.7%	60°
Mini Pretzel	$\frac{1.5}{12}$	12.5%	45°
Chocolate Chips	$\frac{1}{12}$	8.3%	30°
Butterscotch	$\frac{1}{12}$	8.3%	30°



2. Exchange **only** your completed circle graph with your partner and use your partner's completed circle graph to answer the following questions.
   a. Which ingredient was used most in your partner's recipe?
   Square wheat cereal and rice cereal
   b. Which ingredient was used least?
   Corn chips and cheddar crackers
   c. Record your observations about any similarities or differences between your graph and your partner's graph.
   Answers will vary. We both have mini pretzels in our recipe, which seems to take up a similar portion of the recipe for both of us. Otherwise, our ingredients and portions are very different.
   d. Complete the table by using your partner's graph to calculate the number of cups necessary for each ingredient.

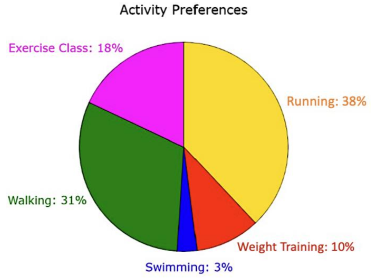
Partner's Recipe Party Ranch Mix
Total Number of Cups Used 13

Ingredient	Number of Cups
Square Wheat Cereal	4
Square Rice Cereal	4
Mini Pretzels	2
Corn Chips	1.5
Cheddar Crackers	1.5

   e. Explain the process you used to determine the number of cups for each ingredient in your partner's recipe.

7.4.4 Your Turn!

1. A health club surveyed 500 of its members to determine which activities they come to the club to participate in. Use the graph to answer the questions.



   a. Describe how to determine which two activities approximately $\frac{2}{3}$ of the members participate in.
   $\frac{2}{3}$ is approximately 67% and the two activities that have a sum closest to $\frac{2}{3}$ are Running and Walking.

   b. How many members come to the club for swimming?
   $0.03 \times 500 = 15$

   c. How many club members participate in an exercise class?
   $0.18 \times 500 = 90$

7.4.4 YOUR TURN!

QUESTIONING

If students finish early, consider asking them to determine how many club members participated in each of the remaining categories represented on the graph.

7.4.5 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of interpreting data in a circle graph.

   Answers will vary.

2. Complete the statement.

   I chose this emoji because

   Answers will vary.

7.4.5 WRAP-UP: REFLECTION

DIFFERENTIATE INSTRUCTION

Consider providing a word wall or anchor chart to provide scaffolding for some of the concepts that were covered in this lesson.